

David Merrell

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My Goals

I seek a **machine learning** research or engineering position, starting Winter 2026.
I want to harness my expertise in **ML, Bayesian modeling, and software engineering** to solve industry problems.
I have recently worked in computational biology, but am open to other sectors as well.
A culture of **software engineering excellence** is non-negotiable.

Employment

Broad Institute; Getz Lab

July 2023–Sep 2025

Computational Biologist II. I led computational efforts for 6 different cancer genomics projects. This entailed:

- preprocessing and quality control for WGS, WES, scRNA-seq, and bulk RNA-seq data;
- somatic and germline variant calling from genomics;
- finding associations between -omics and clinical variables.

DataChat

Summer 2019

ML Engineer Intern. I built machine learning backend infrastructure for a chatbot-based data analysis product.

RAND Corporation

2014–2016

Research assistant. I wrote simulations and ran analyses of the DoD's transportation capabilities and fuel usage.

Sandia National Laboratories

Summer 2013; Summer 2014

Intern. I built material heterogeneity models for the ALEGRA shock and multiphysics simulation code, in C++.

Pacific Northwest National Laboratory

Summer 2012

Intern. I quantified the performance of a computer vision model for the NIFFTE nuclear fission experiment.

Education

PhD Computer Sciences

2018–2023

University of Wisconsin–Madison. I built probabilistic models for -omics data and biological pathways.
Advised by Tony Gitter.

MS Computer Sciences

2016–2018

University of Wisconsin–Madison. I researched techniques for probabilistic and logical inference.
Advised by Aws Albarghouthi and Loris D'Antoni

BS Mathematics

2014

Brigham Young University. Minor in Physics. 3.90 GPA. *Cum Laude*.

Software

PathMatFac.jl

Julia

A matrix factorization model for multiomic data. Uses curated gene sets to regularize the factorization.

Sparse Signaling Pathway Sampling (SSPS)

Julia; Snakemake

A method that infers signaling pathway structure from time series phosphoproteomic data.

TrialMDP

C++; R

An algorithm that designs clinical trials with optimal response-adaptive randomization.

TCGA Pipeline

Python; Snakemake

A pipeline that downloads multiomic TCGA data and merges them into an HDF5 suitable for machine learning.

Publications

J. Ghannam, . . . , D. Merrell, "Tumor Transcriptional Subtype Is a Key Axis of Immune Checkpoint Blockade Response in Glioblastoma". *Nature Cancer*. Under review.

H. Parsons, . . . , D. Merrell, "Detection of Heterogeneous Resistance Mechanisms to Tyrosine Kinase Inhibitors from Cell-free DNA". *Cell Genomics*. Sep 2025.

B. Anderson, . . . , D. Merrell, "Barcoded viral tracing identifies immunosuppressive astrocyte-glioma interactions". *Nature*. June 2025.

D. Merrell, T. Chandereng, Y. Park. "A Markov Decision Process for Response-Adaptive Randomization in Clinical Trials". *Computational Statistics and Data Analysis*. Feb 2023.

D. Merrell, A. Gitter. "Inferring Signaling Pathways with Probabilistic Programming". *European Conference on Computational Biology*. 2020.

D. Merrell, A. Albarghouthi, L. D'Antoni. "Weighted Model Integration with Orthogonal Transformations". *International Joint Conference on Artificial Intelligence*. 2017.

Skills

Programming languages: Python; Julia; Rust; C++; R; MATLAB

Packages & libraries: pytorch, jax, scikit-learn, numpy, pandas, matplotlib, seaborn, statsmodels, scanpy

Other tools: git, Snakemake, Docker, Google Cloud Platform, SQL, $\text{\LaTeX}$, jupyter, GATK, samtools, Cursor

-Omic Data modalities: Whole genomes; scRNA-seq; whole exomes; bulk RNA-seq

Non-technical: Writing, public speaking, event planning, teaching

Service and Leadership

Broad Cancer Program Bootcamp

April 2025

Teacher. Taught a week-long python course to early-career cancer researchers.

UW-Madison Student ACM chapter (SACM)

2018

President. Restructured the organization. Improved the division of labor. Recruited 25 student officers with clearly defined responsibilities. Increased budget by 50% through fundraising. Planned and executed numerous social and professional events.

Eagle Scout

2007 – present

Personal Interests

Running, biking, swimming, hiking, reading, writing, personal coding projects